15 References — Verification Links

Each reference below includes a direct DOI, publisher, or bibliographic link so we can check its authenticity and details.

1. Kumar, V., Mukil, S. & Naveen, P. Modeling and evaluation of the permeate volume in membrane desalination processes using machine-learning techniques. Digital Chemical Engineering 11, 100154 (2024).

https://doi.org/10.1016/j.dche.2024.100154

2. Alanazi, M. & Alharby, T. N. Computational evaluation using machine learning for analysis of membrane desalination process powered by solar energy. Scientific Reports 15, 36384 (2025).

https://doi.org/10.1038/s41598-025-20291-4

3. Rasmussen, C. E. & Williams, C. K. I. Gaussian Processes for Machine Learning. MIT Press (2006).

https://mitpress.mit.edu/9780262182539/gaussian-processes-for-machine-learning/

4. Efron, B. & Tibshirani, R. J. An Introduction to the Bootstrap. Chapman & Hall/CRC (1993).

https://books.google.com/books?id=gLlpIUxRntoC

5. Chou, J. S. & Truong, D. N. A novel metaheuristic optimizer inspired by behavior of jellyfish in ocean. Applied Mathematics and Computation 389, 125535 (2021).

https://doi.org/10.1016/j.amc.2020.125535

6. Chen, T. & Guestrin, C. XGBoost: a scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (2016), 785–794.

https://doi.org/10.1145/2939672.2939785

7. Breiman, L. Stacked regressions. Machine Learning 24, 49–64 (1996).

https://doi.org/10.1023/A:1018046112532

8. Thajudeen, K. Y., Ahmed, M. M. & Alshehri, S. A. Integration of machine learning and CFD for modeling mass transfer in water treatment using membrane separation process. Scientific Reports 14, 23970 (2024).

https://doi.org/10.1038/s41598-024-74530-1

9. Wei, Y. Development of novel computational models based on artificial intelligence technique to predict liquids mixtures separation via vacuum membrane distillation. Scientific Reports 14, 24121 (2024).

https://doi.org/10.1038/s41598-024-75074-0

10. Ashraf, W. M. & Dua, V. Data information integrated neural network (DINN) algorithm for modelling and interpretation performance analysis for energy systems. Energy AI 16, 100363 (2024).

https://doi.org/10.1016/j.egyai.2024.100363

11. Ashraf, W. M. et al. Machine learning assisted improved desalination pilot system design and experimentation for the circular economy. Journal of Water Process Engineering 63, 105535 (2024).

https://doi.org/10.1016/j.jwpe.2024.105535

12. Shalaby, S. M. et al. Membrane distillation driven by solar energy: a review. Journal of Cleaner Production 366, 132949 (2022).

https://doi.org/10.1016/j.jclepro.2022.132949

13. Van der Bruggen, B. & Vandecasteele, C. Distillation vs. membrane filtration: overview of process evolutions in seawater desalination. Desalination 143, 207–218 (2002).

https://doi.org/10.1016/S0011-9164(02)00247-5

14. Anusha, P. V. et al. Detecting outliers in high dimensional data sets using Z-score methodology. International Journal of Innovative Technology and Exploring Engineering 9(1), 48–53 (2019).

https://doi.org/10.35940/ijitee.A3910.119119

15. Curran-Everett, D. Explorations in statistics: the log transformation. Advances in Physiology Education 42(2), 343–347 (2018).

https://doi.org/10.1152/advan.00018.2018